Markscheme

Mathematics: analysis and approaches
Practice question bank

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91. (a)

Consider the statement: "$n^2 + n + 1$ is odd for every integer n satisfying $1 \leq n \leq 4$."

Verify the statement is true by checking each case (proof by exhaustion), and hence find the sum of the four values of $n^2 + n + 1$ obtained.

check each case: $n = 1 : 3, n = 2 : 7, n = 3 : 13, n = 4 : 21$ *(M1)* **A1**

each value is odd, so the statement holds for $1 \leq n \leq 4$ **R1**

sum $= 3 + 7 + 13 + 21$ *(M1)*

$= 44$ **A1**

[5 marks]